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Debugger: Taking Control

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Starting Off Right

- Assumed: SDK, knowing Palm OS® (at least a little), documentation, C programming…
- Essential: Palm OS Emulator, Debug roms
- Helpful additions: PalmDebugger, PalmReporter, knowing 68k assembly, books, the Palm OS Source code
- Information sources: PalmOS.com's KnowledgeBase, developer forums

The Basics

- Always develop on debug ROMs, the newer the better
- Test on various ROMs and hardware configurations
  - Perhaps 10% on release roms and real hardware
- Run Gremlin Hordes
  - (They work at night, for free!)
- Use the APIs, don't build on shifting sand… (unless you really want to)

Saving Time With POSE

- Save .PSF files with good starting configurations, or put PRCs in the AutoLoad directory
- Set the hard buttons to launch your application or do something special
- Add debugging code, which only runs on Pose, or uses HostControl to output information when given special input

Saving Time With CodeWarrior

- (For r6, make sure you've got the 3.5 SDK update plus the debugger patch)
- Turn on the browser, so Palm OS® functions, constants, etc. are light blue
- Sort function popup, add separator lines
- Add default data into the application itself
  - Perhaps that's another target
- Use the Register window, call functions twice, change variables by hand

Finding Information

- Use the frames version of the SDK docs
- Set up “File Sets” for the Find dialog, or your favorite text searching app which can show hits in multiple files simultaneously

Saving Time With the Debugger

- Use it to convert between hex and decimal, and for doing math
- Set aliases of common commands
  - See the ones that are in the 4.0 SDK
- Just hit return to repeat last command, and/or move the cursor and re-execute
- (More walkthroughs to come later)

The Hard Things to Debug

- Uninitialized variables
- Bad pointers and Memory stomps, and things that only show up later on
- Problems appearing inside OS routines

- Tracing event flow, or finding out what the gremlin was doing
- Overrunning the stack
- Communications

PalmDebugger Crash Course

- att, atb, br, sc, il pc-20, dm, reset
  - Remember about il pc-22, and sc 20
  - Good places to use with “atb”: SysAppLaunch, EvtGetEvent, FtrGet, MemHandleNew, FrmDrawForm, ExgAccept
- s, t, pc=pc+2, hd 0, ss
- dir, import, opened, listresources, listrecords

Debugging Confusions

- “br” versus “atb” versus “ss”
- Stepping over a system trap can be confusing
- “t” versus “s” in PalmDebugger versus CodeWarrior, and F10/F11
- Heap formats and debugger versions

Walkthrough: Getting a Gremlin

- By using POSE's traces
- By using PalmDebugger

Debugging on a Device

- Shortcut-tap-tap-1 and 2
  - Flashing square or “thin green line”, open serial port…
- Connecting and hitting “g” before doing testing
- Use a rechargeable device… or else you might experiment with a DC power supply set to 3 v
  - Which might break the warranty; you’re on your own…
- sb 101 81 (unsupported HotSync button hack)

The Stack

- register A7 is the stack pointer, also called SP
- register A6 usually points to parameters
- “sc” uses A6 to follow stack frames back
  - Use “sc a6 20” to see previous 20 callers
- hd 0 to see size and location of stack
- Try “ss” to pinpoint where it overflows
- Compare addresses of two local variables to see how much stack space is being used
- Remember, stack sizes vary

Debugging Communications

- For Exchange Manager:
  - Use “Local” IR mode, or “Transfer” on 4.0
  - Use sb 101 81 hack on a real device
- Write status to a log (Memo, or your own)
- Write a stdio application
- Use PalmReporter
- Draw on the screen
  - LED-style for status
  - “Piano roll” style dots
- Use Palm OS® Emulator for TCP/IP

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Q&A


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